

Handling Missing Values in STATA

http://www.ats.ucla.edu/stat/stata/faq/nummiss_stata.htm

1. Number of missing values vs. number of non missing values

The first thing we are going to do is determine which variables have a lot of missing values. We have created a small Stata program called **tabmiss** that counts the number of missing values in both numeric and character variables. You can download **tabmiss** from within Stata by typing

```
findit tabmiss
```

(see [How can I used the findit command to search for programs and get additional help?](#) for more information about using **findit**).

```
tabmiss landval improval totval salepric saltoapr
```

2. Obtaining the distribution of the number of missing values per observation

We can also look at the distribution of missing values across observations. For example variable **nmis** created below is the number of missing values across each observation. Looking at its frequency table we know that there are four observations with no missing values, nine observations with one missing values, one observation with two missing values and one observation with three missing values. If we are willing to substitute one missing value per observation, we will be able to reclaim nine observations back to get a valid data set that is $13/15=87\%$ of the size of the original one. The function **rmiss2()** used here is an extension to the **egen** function **rmiss()**. It counts the number of missing values in the varlist. **rmiss2()** accepts both string and numeric variables. Stata's **rmiss()** accepts only numeric variables. You can download **rmiss2()** over the internet from within Stata by typing **findit rmiss2** (see [How can I used the findit command to search for programs and get additional help?](#) for more information about using **findit**). Now let's create our new variable.

```
egen nmis=rmiss2( landval improval totval salepric saltoapr)
```

```
tab nmis
```

3. Patterns of missing values

We can also look at the patterns of missing values. We can recode each variable into a dummy variable such that 1 is missing and 0 is nonmissing. Then we use the **collapse** command to compute the frequency for each pattern of missing data

```
egen land1=rmiss2(landval)
egen impr1=rmiss2(improval)
egen totv1=rmiss2(totval)
egen sale1=rmiss2(salepric)
egen salt1=rmiss2(saltoapr)
gen id=_n
collapse (count) freq=id, by (land1 impr1 sale1 totv1 salt1)
```

IMPUTING Missings in STATA

IMPUTE: Also use STATA 'impute' command, as an ado file, (page 217-221 in the STATA 9 (D) Data Management Manual:

Examples

```
. impute ln_rtl jantemp precip ln_inc medage hhsize,
gen(i_ln_rtl)
. regress ln_eat i_ln_rtl jantemp precip ln_inc medage hhsize
. factor mpg-foreign
. score f1 f2
. impute f1 mpg-foreign, gen(i_f1)
. impute f2 mpg-foreign, gen(i_f2)
. regress price i_f1 i_f2
```

To impute all . missings (and so not the extended missing values .a, ..., .z) using all valid observations,

```
. impute edu xvars if edu==., gen(iedu) all
```

Note: When imputed missings are included the regression coefficient a bias may result which is the ratio of the mean of the variable including imputed values to the variance of the variable without imputed values; Bias is usually towards zero, this results in a marginally underestimated regression coefficient.

For different discussions on imputation see little & Rubin (2002) and (Cohen, Cohen, West, & Aiken, 2003). Also see Manual: [D] impute

IPOLATE: Sometimes we can linearly interpolate (or extrapolate) the missings from existing data; e.g. if we are missing shipment numbers for a product that has reasonably constant demand, like newspapers. In this case use the STATA ipolate command.

Examples

```
. ipolate y x, gen(yprime)
. ipolate y x, gen(yprime) epolate
. by magazine: ipolate circ year, gen(icrc)
```

Also see Manual: [D] ipolate

Frijters Method: creates a dummy variable to replace the missings. The advantage of this method is that it maximises sample size by substituting the missing values with a dummy value that stops STATA from kicking out an observation that may only have a missing on one of its many variables. How it works:

```
*** Professor Frijters method for replacing missing values in a
variable. It assigns a value to the missing so that the number of
* observations can be maximised. Useful for regressions but don't use
the replaced values for descriptive statistics, use the raw data.
** Generate an average of the variable
*
egen avgvarname = mean(varname)
*
** Generate a Dummy to identify missing observations
*
generate missingvarname=0
*
** Dummy = 1 for the observations that have the HILDA missing value
such as "." Note: HILDA, assigns different missing values by
variable
*
replace missingvarname=1 if varname=.
*
** Replace missing observations with the average of the variable; the
sign of the variables not being replaced are reversed
* mvarname is the name for a variable that has the missings
corrected. Use this variable name in the regressions.
*
gen mvarname=varname*(missingvarname-1)+ (missingvarname)*
(avgvarname*-1)
*
** The following line corrects for the reversed sign of mvarname and
changes it back to the sign of the varname
*
replace mvarname=mvarname*-1
*
* notes: control for the effect of the inclusion of the average value
in the missings by including avgvarname in regressions.
*
* notes: Remember to use the missing corrected variable (mvarname) in
the regressions.
* notes: Don't use the averaged observations in scatterplot and
histograms; the plots will look silly with a big lump in the middle
```

Inputs from Stephen Cox:

There is also good evidence that single imputation methods which are regression based are not particularly good (e.g., the impute command in Stata; the regression method in SPSS). This is one area where Stata is a little behind (although ICE solves that). Paul's method is not an approach that is accepted these days either. (i think he is essentially doing a mean substitution and then adding in a missing dummy variable, but i haven't looked closely).

see, e.g.,

[Indicator and Stratification Methods for Missing Explanatory Variables in Multiple Linear Regression](#)

Michael P. Jones

Journal of the American Statistical Association, Vol. 91, No. 433. (Mar., 1996), pp. 222-230.

Stable URL: [http://links.jstor.org/sici?sici=0162-](http://links.jstor.org/sici?sici=0162-1459%28199603%2991%3A433%3C222%3AIASMFM%3E2.0.CO%3B2-C)

[1459%28199603%2991%3A433%3C222%3AIASMFM%3E2.0.CO%3B2-C](http://links.jstor.org/sici?sici=0162-1459%28199603%2991%3A433%3C222%3AIASMFM%3E2.0.CO%3B2-C)

Abstract: The statistical literature and folklore contain many methods for handling missing explanatory variable data in multiple linear regression. One such approach is to incorporate into the regression model an indicator variable for whether an explanatory variable is observed. Another approach is to stratify the model based on the range of values for an explanatory variable, with a separate stratum for those individuals in which the explanatory variable is missing. For a least squares regression analysis using either of these two missing-data approaches, the exact biases of the estimators for the regression coefficients and the residual variance are derived and reported. The complete-case analysis, in which individuals with any missing data are omitted, is also investigated theoretically and is found to be free of bias in many situations, though often wasteful of information. A numerical evaluation of the bias of two missing-indicator methods and the complete-case analysis is reported. The missing-indicator methods show unacceptably large biases in practical situations and are not advisable in general.

There is a multiple imputation (MI) program written for Stata, called ICE.

<http://www.ats.ucla.edu/stat/stata/library/ice.htm>

There is very good statistical evidence that MI works as well as ML estimates, but MI is more fiddly. However, ICE makes it somewhat easier (although quite some steps still). I have not seen any formal evaluation of ICE versus the other MI programs that are floating around, however. (E.g., AMOS now comes with such a procedure...). My unsolicited recommendation would be to at least look at ICE.

Cheers, Stephen Cox

Using SPSS MVA to correct for Missings

Load data file (*.dta) into STATA.

Run tabmiss to get a frequency table of missings by variable, then save *.dta.

In SPSS 15, File/open/data & select STATA file type then select and load the *.dta file that requires missings replaced.

For large data sets SPSS may run out of memory, run a program to SET the workspace

SET workspace = 6149 (note: 6149 kbytes is the default and needs to be increased to fit the dataset size)

The following SPSS syntax program is in:

[H:\Tony Beatton\IF49\Analysis\Handling missing data\set SPSS workspace.sps](#)

* A short program to SET the SPSS Workspace to allow for large data sets like HILDA.

* .

cd "H:\Tony Beatton\IF49\Analysis\Handling missing data"

set workspace = 38764.

* End

Now that the data are in SPSS, check the variable view to ensure that SPSS transformed the variable definitions correctly. View the data and spot check that missings are as expected.

Now replace the missings using the maximum likelihood imputation technique with EM estimation algorithm in SPSS 9 (this is based on an assumption that an inclusive strategy is better than an exclusive model). We use all items to determine missing data. The procedure uses all items to determine missings.

Procedure:

Menubar: Analyse/Missing Value Analysis

Select 'use all variables' and move to Quantitative

Can put categorical variables into Quantitative as well. However, the procedure doesn't handle string variables.

Select 'Variables' radio button

'Select all variables' button and move all variable into 'both' control boxes

Continue

Select 'EM' radio button

Select 'Normal' radio button

Select 'save as data' radio button

File: input filename

Set maximum iterations to 100

Continue

OK

The MVA procedure is run and the MVA report is output. Use it to check for the nature of missings, and the effect of replacing the missings

Save the missings-corrected data file as STATA Version 8 SE (*.dta)

In STATA/SE for large data sets, load the missings-corrected *.dta file. Run tabmiss to check for missings occurrences, there should be none.